STATS 200A Discussion 4

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Theorem 4.1 (Monotone Convergence Theorem)

$$f_n \geq 0, f_n \uparrow f = \begin{cases} f_n \leq f_{n+1} \\ \lim_{n\to\infty} f_n(x) \overset{p}{=} f(x) \quad \forall x \in \mathbb{R} \end{cases}$$

Remark.

$$\begin{aligned}
\lim_{n\to\infty} \int_{\mathbb{R}} f_n \, d\mu & \\
&= \int_{\mathbb{R}} \lim_{n\to\infty} f_n \, d\mu \\
&= \int f \, d\mu \\
f_n(x) &= n \cdot \mathbf{1}_{x\in[0,\frac{1}{n}]} \\
f &= 0 \\
\int f \, d\mu &= 0 \\
\lim_{n\to\infty} \int f_n \, d\mu &= 1 \\
\forall n \quad \int f_n \, d\mu &= n \cdot \int \mathbf{1}_{[0,\frac{1}{n}]} \, d\mu \\
&= n \cdot \mu\left(\left[0, \frac{1}{n}\right]\right) \\
&= n \cdot \frac{1}{n} = 1
\end{aligned}$$

Proof. Step 1:

$$\lim_{n\to\infty} \int f_n \, d\mu$$

exists?

$$\begin{aligned}
f_n \leq f_{n+1} \Rightarrow \int f_n \, d\mu \leq \int f_{n+1} \, d\mu & \\
\limsup \int f_n > \liminf \int f_n & \\
\varepsilon > 0 \varepsilon = \limsup \int f_n - \liminf \int f_n & \\
\exists N \quad \int f_N \geq \limsup \int f_n - \frac{\varepsilon}{4} &
\end{aligned}$$

$$\exists M \quad \int f_M \geq \liminf \int f_n - \frac{\varepsilon}{4}$$
$$\limsup x_n = \overline{x}$$
$$\liminf x_n = \underline{x}$$
$$x_{n_k} \to \overline{x}$$
$$x_{n_{k'}} \to \underline{x}$$
$$\overline{x} - \underline{x} = \varepsilon > 0$$
$$\exists N : |x_N - \overline{x}| \leq \frac{\varepsilon}{4}$$
$$\exists M : |x_M - \underline{x}| \leq \frac{\varepsilon}{4}$$
$$M \geq N \Rightarrow x_M \geq x_N$$
$$L = \int f \, d\mu$$

WTS: $L = \lim_{n\to\infty} \int f_n \, d\mu$

$$L \gtreqless \lim_{n\to\infty} \int f_n \, d\mu$$
$$\forall n : f_n \leq f : \int f_n \leq \int f$$
$$\lim \int f_n \, d\mu \leq \int f \, d\mu = L$$
$$f = \mathbf{1}_{x\in[a,b]}$$
$$f = \sum_{i=1}^{k} \alpha_i \cdot \mathbf{1}_{x\in A_i}$$
$$f \geq 0 : \int f \, d\mu := \sup_{s:\text{simple function}} \left\{ \int s d\mu = s \leq f \right\}$$
$$s \leq f \to \int s \leq \int f$$
$$0 < c < f E_n := \{x : f_n(x) \geq c \cdot s(x)\}$$
$$E : \{s \leq f\} = \mathbb{R}$$
$$E_1 \subseteq E_2 \subseteq E_3 \cdots E = \mathbb{R} = \lim_{n\to\infty} E_n$$
$$E_1 \subseteq E_2 \subseteq \cdots$$
$$E_i \uparrow E$$
$$\lim_{n\to\infty} \mu(E_i) = \mu(E) = \mu\left(\bigcup_{i=1}^{\infty} E_i\right)$$
$$\lim_{n\to\infty} \mu\left(E_n\right) = \mu(E)$$
$$\lim_{n\to\infty} \mu\left(E_n^c\right) = \mu\left(E^c\right) = 0$$
$$\int_{\mathbb{R}} f_n \, d\mu = \int_{E_n} f_n \, d\mu + \int_{E_n^c} f_n \, d\mu$$
$$\geq \int_{E_n} s \, d\mu + \int_{E_n^c} s \, d\mu$$
$$\forall s \leq f : \lim_{n\to\infty} \int_{\mathbb{R}} f_n \geq \int s \, d\mu$$

$$\Rightarrow \lim_{n\to\infty} \int f_n \, d\mu \geq \underbrace{\sup_{s\leq f} \int s \, d\mu}_{\int f \, d\mu} = L$$

for $s = \sup_{s'} \{s \leq f\}$ ■